

TERMINOLOGY

Definitions

- Cell
 - Basic structural unit of all tissue
- Organelles
 - Intracellular heterogeneous interactive substructures

OVERVIEW OF CELL ORGANIZATION

Membrane

- Major unit of organization
- Membranes appear as single linear structure by light microscopy and low-magnification electron microscopy
- Phospholipid bilayer
 - Hydrophilic surface, hydrophobic center with interspersed variable-sized protein globules

Membrane-Bound Organelles

- Nucleus
- Plasma (cell) membrane
- Endoplasmic reticulum (ER)
 - Rough and smooth types
- Golgi apparatus
- Mitochondria
- Lysosomes
- Cell membrane specializations
 - Microvilli, pinocytotic vesicles
- Various types of secretory/neuroendocrine granules

Nonmembrane-Bound Organelles

- Nuclear chromatin, nucleolus
- Intracytoplasmic contractile and anchoring filaments
- Cytoskeletal tubules
- Free ribosomes
- Glycogen, fat globules
- Desmosomes, attachment plaques
- Centrioles, basal bodies

Extracellular Elements

- Basal lamina, collagen, matrix proteoglycans

NUCLEUS

Chromatin

- Electron-dense aggregates representing DNA
 - Heterochromatin: Clumped, electron-dense DNA, biologically inactive
 - Euchromatin: Dispersed DNA, biologically active
- Individual chromosomes not apparent without cell culture

Chromosomes

- 46 total chromosomes per human cell
 - 22 pairs of identical chromosomes (or autosomes) with either 2 X chromosomes for females or 1 X and 1 Y chromosome for males
- 2 chromatids join together at centromere to form chromosome
- Telomeres are located at end of each chromosome

Nucleolus

- Focus of variable prominence in nucleus
- Filamentous and granular nucleolonema composed of histones, ribonucleic acid (RNA), and regulatory cell cycle proteins

Cytoplasmic Interactions

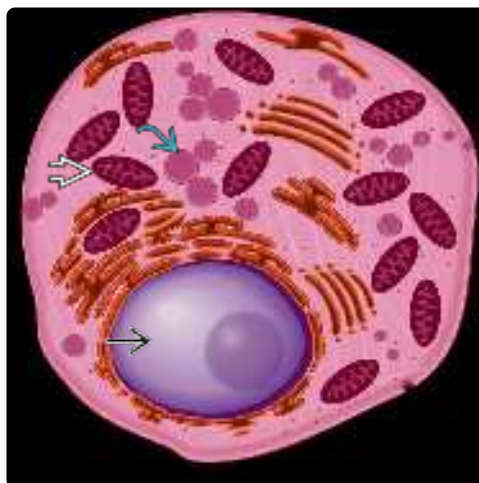
- Via
 - Nuclear pores
 - Nuclear blebs
 - Cisternal space contiguous with ER
 - Dissolution of nuclear membrane during mitosis

CYTOPLASM

Endoplasmic Reticulum

- Definition
 - Parallel arrangement of membranes adjacent to and contiguous with nucleus
- Rough ER (rER)
 - Ribosomes are free, small granular aggregates of cytoplasmic RNA, which, when attached to ER membranes, constitute rER

Cellular Components



Cellular Organization



(Left) This depiction of organelle distribution within a cell shows a basally placed nucleus , mitochondria , endoplasmic reticulum, and variably sized granules  (mucin or other proteins). (Right) Endocervical cells from a Pap smear show typical cellular organization (basal nucleus, variable chromatin clumping, terminal bar  with cilia, granular cytoplasm, and vacuoles, Golgi  with mucin droplet formation).